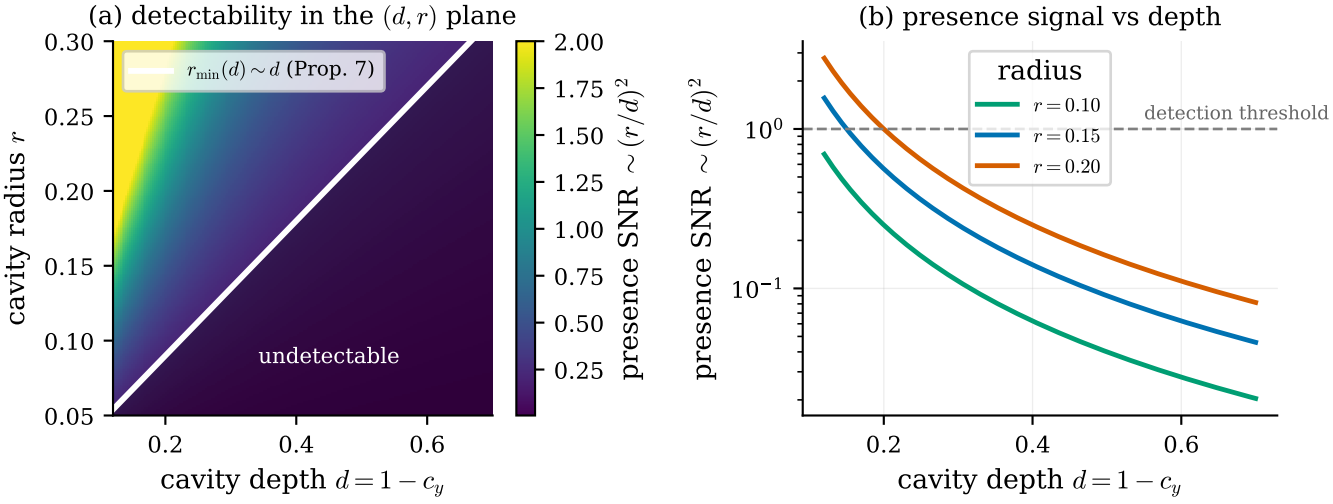

(a) detectability in the (d, r) plane
$r_{\min}(d) \sim d$ (Prop. 7)
undetectable
cavity radius r
cavity depth $d = 1 - c_y$
presence SNR $\sim (r/d)^2$
(b) presence signal vs depth
radius
$r = 0.10$
$r = 0.15$
$r = 0.20$
detection threshold
presence SNR $\sim (r/d)^2$
cavity depth $d = 1 - c_y$